

	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	09 Apr 2015	First Release	CHIN CHEE YAN
01	2 Oct 2016	Template update & Content update	ONG KEAN SHENG

	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

Content

Low Magnification (19um) Tube Height Adjustment Procedure.....	3
High MAG Tube Height Adjustment Procedure.....	15

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

Low Magnification (19um) Tube Height Adjustment Procedure

1. Go to path C:\Program Files\AXI System\v810\<REV>\config and open software.config with notepad format.
2. Search for “saveAllConfirmationImages”, set indicator value to “true” and save config. Show as Figure 1.

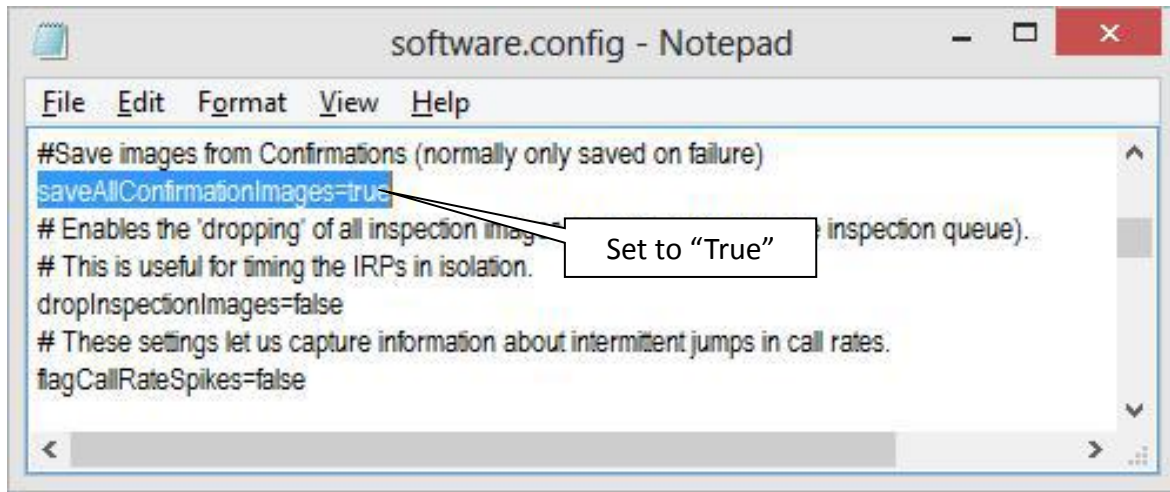


Figure 1 Software Config

3. Quit V810 GUI and restart V810 GUI as usual.
4. Perform 1 loop full CDnA task to obtain latest log file. Go to Confirmation, Diagnostic and Adjustment tab and as show in Figure 2.

Services → Confirmation, Diagnostic and Adjustment → Scheduled Tasks → Run 1 loop

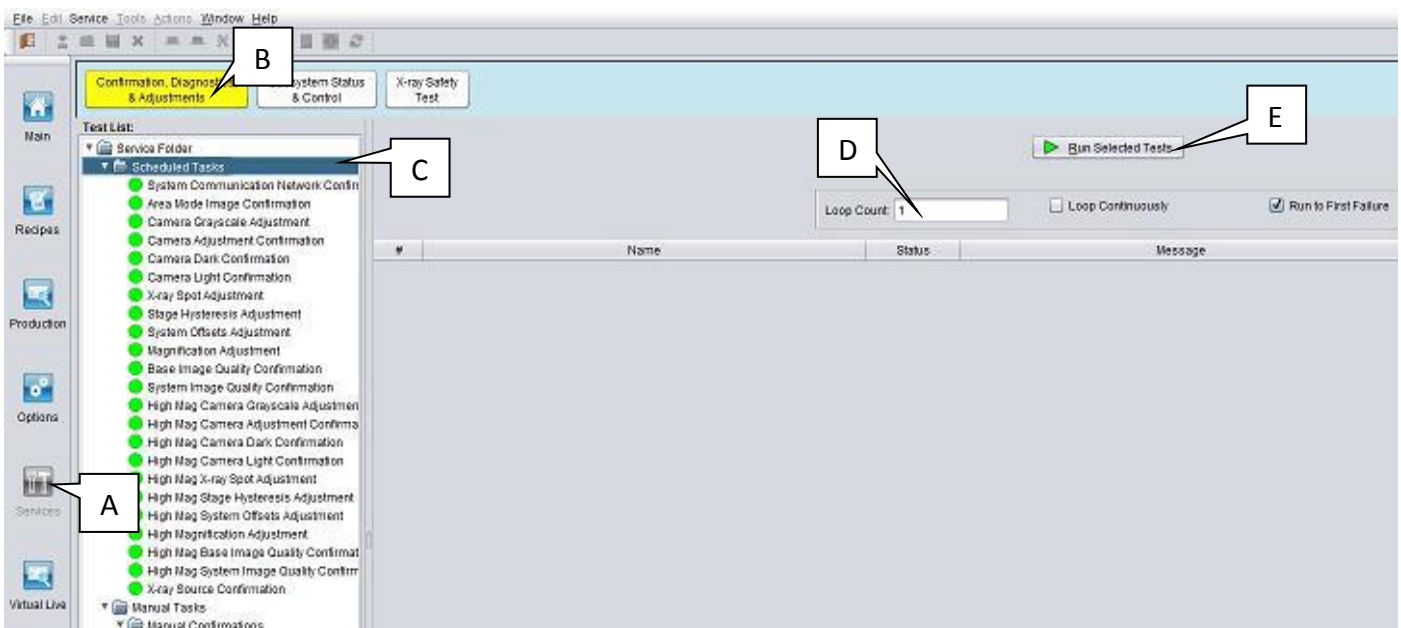


Figure 2 CD & A

	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

5. Once done perform full CDnA, go to path C:\Program Files\AXI System\v810\<REV>\log\calib\Magnification, check for latest low magnification log file. Sample as Figure 3. (# File name indicate as *magnificationYYYY.MM.DD.HH.MM.SS.XXX.log*)

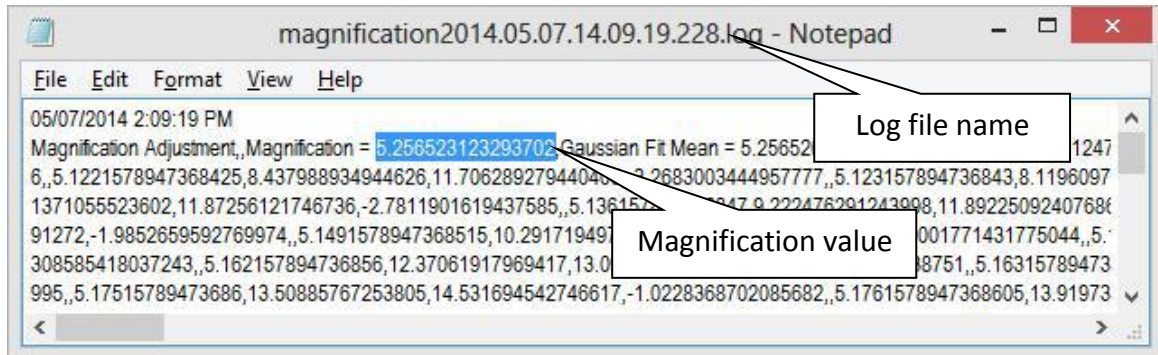


Figure 3 Magnification Log

6. From the latest log file, calculate the required adjustment mils need to perform at micrometer with formula below.

$$[A = (5.25 - B) / 0.004]$$

A = mils required

B = latest log files

1 turn of Micrometer = 40 mils

If calculation result is + value, mean you need to lower tube Z position

If calculation result is - value, mean you need to higher tube Z position

Sample

If latest log file is 5.15

$$A = (5.25 - 5.15) / 0.004$$

$$A = 25 \text{ mil}$$

Require to lower down tube by turn 25mil to achieve 5.25 tube heights.

	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

7. Go to Digital I/O tab and initialize as show in Figure 4.

Services → Subsystem Status & Control → Digital I/O → Initialize

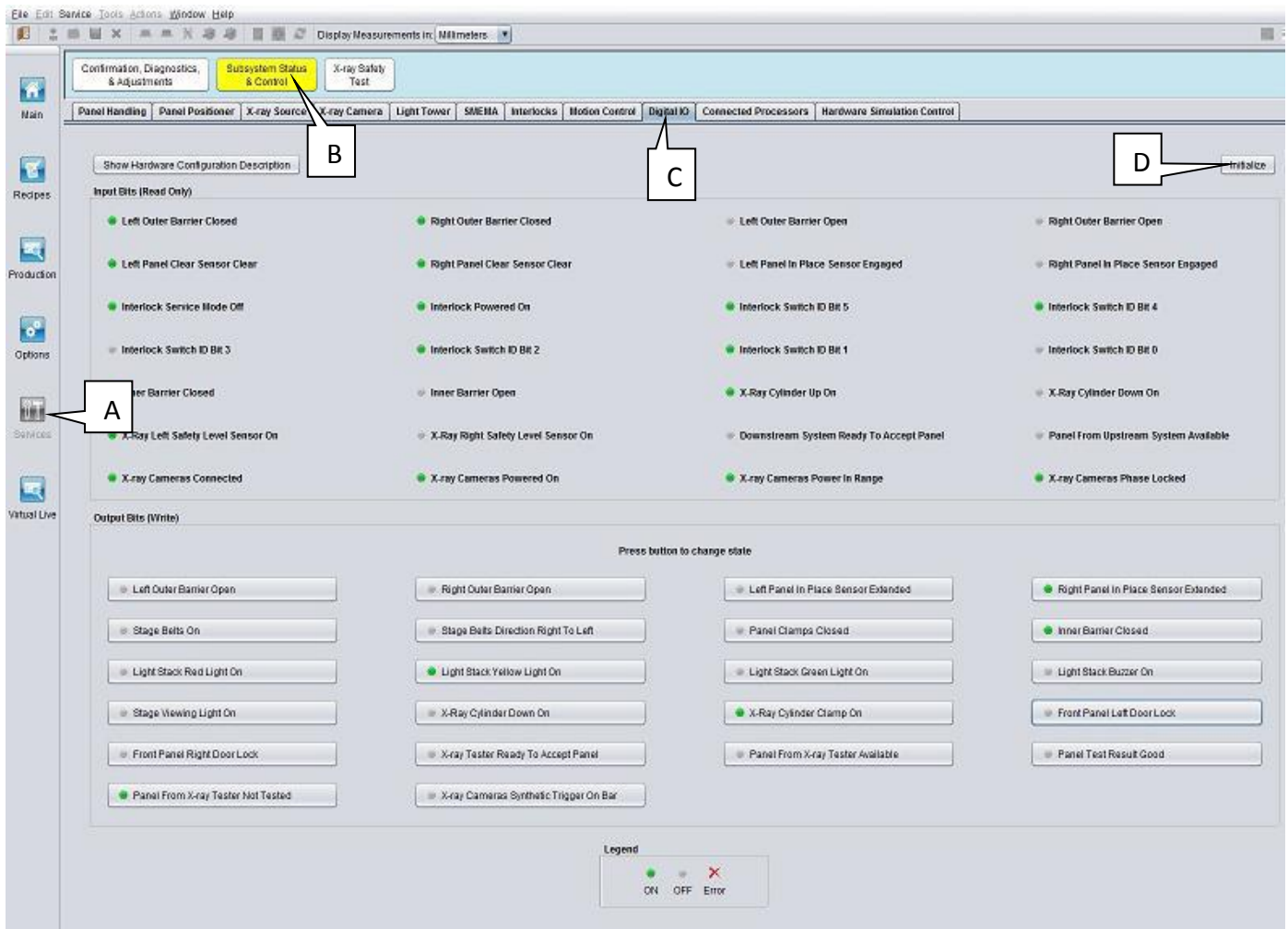


Figure 4 Digital IO

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

8. Go to Panel Handling tab and initialize as show in Figure 5. Change display measurement unit to inch and set AWA rail width to 18 inch. Trigger Inner Barrier in Open position, tube cylinder to “Up” position and Clamp in Open position.

Services → Subsystem Status & Control → Panel Handling → display unit to Inch → Initialize → 18 inch width → Inner Barrier Open Position → X-Ray Cylinder Up Position

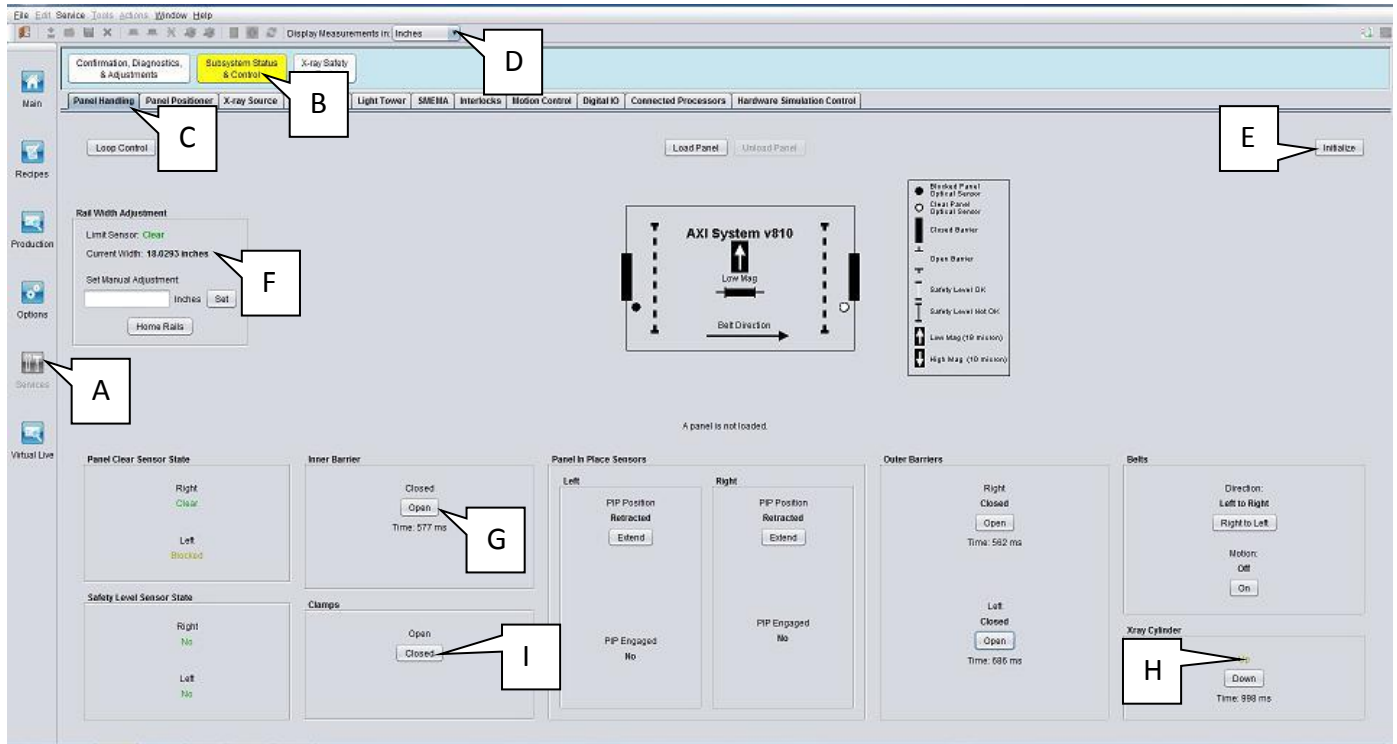


Figure 5 Panel Handling

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

9. Go to X-Ray Source tab turn off the X-Ray voltage as Figure 6. Then turn off X-Ray Key at Operator Console and open the machine Front Access Door.

Services → Subsystem Status & Control → X-Ray Source → Turn Off Current & Voltage

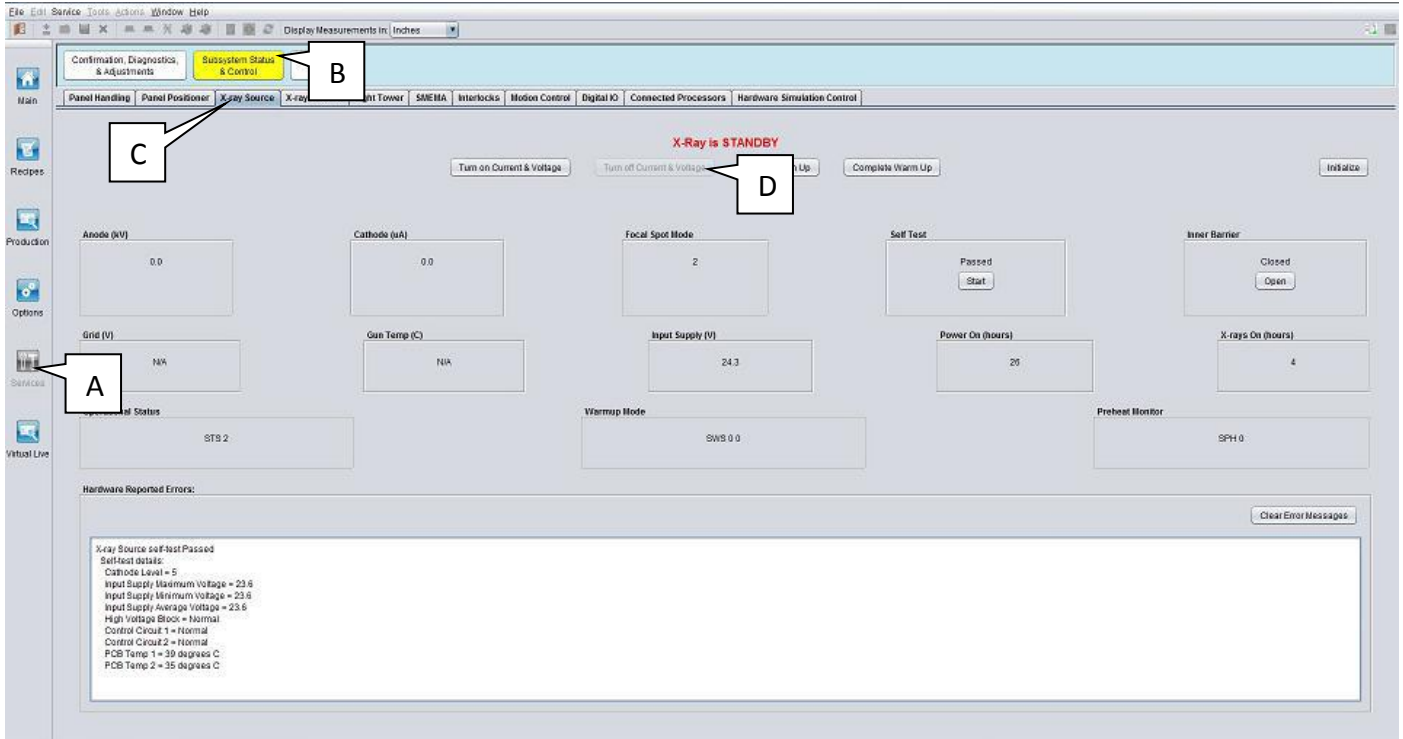


Figure 6 X ray Source

10. Remove the U-Shield Front Cover with loosen 4 x M5 nuts as Figure 7. Left is High Mag hard stop and Right is Low Mag hard stop.

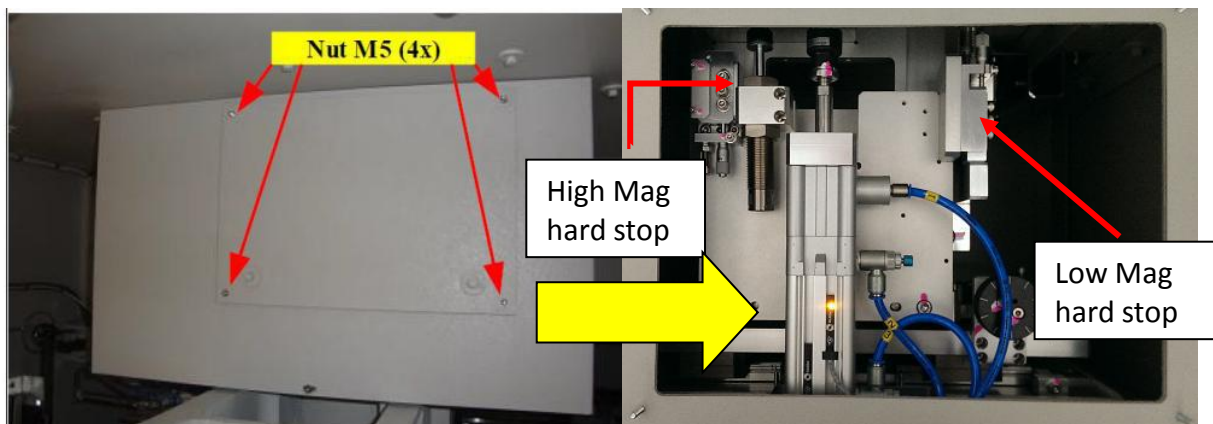


Figure 7 U-Shield Front Cover

	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

11. Push the XY stage Y axis toward to middle and ensure cylinder down position will not have any obstacle especially with the Clamper as Figure 8.

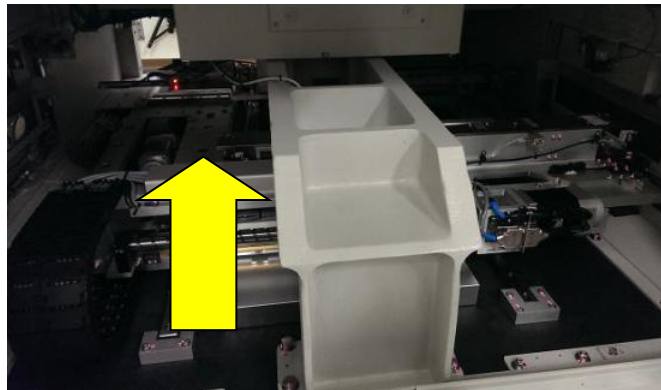


Figure 8 XY Stage movement

12. Log Mag Hard Stop and screws show as Figure 9.

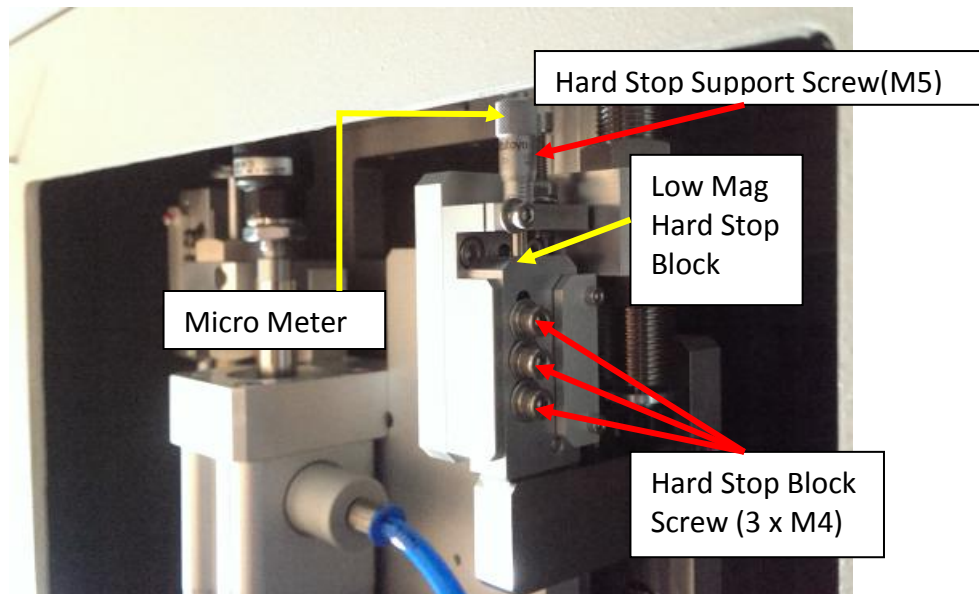


Figure 9 (New Rev)

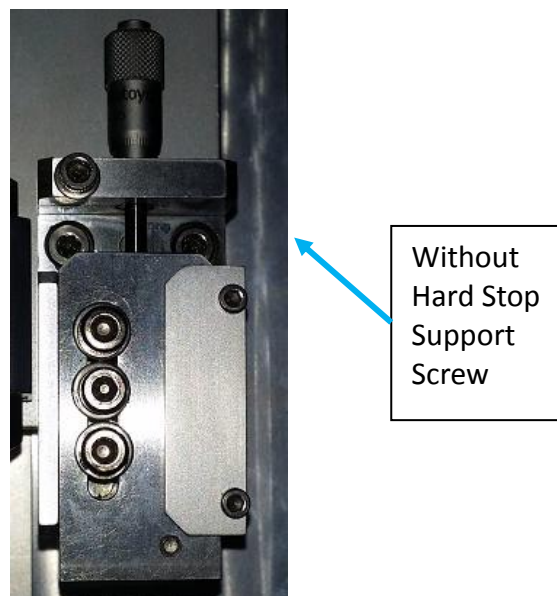


Figure 10 (Old Rev)

	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

Remark,

If machine assembly with VM Old Rev as Figure 10, the difference will be no assembly Hard Stop Support Screw (M5), tube adjustment will be the same method.

13. **Remark, If wish to lower tube Z for Low Mag adjustment; need to trigger tube in Down position 1st.**

Go to Panel Handling tab, trigger tube cylinder to “Down” position 1st as Figure 11.

Services → Subsystem Status & Control → Panel Handling → Inner Barrier Open Position → X-Ray Cylinder Down Position

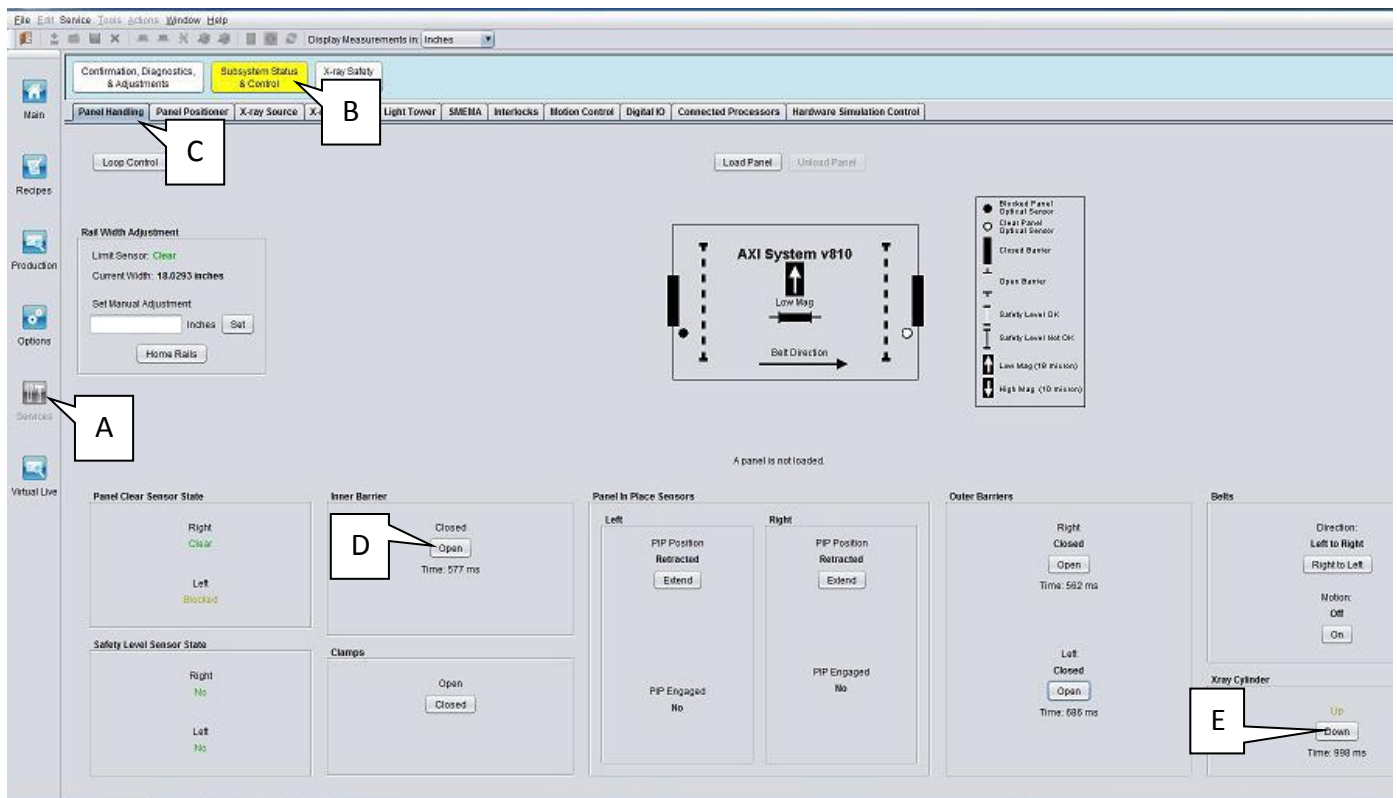


Figure 11 Panel Handling

	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

14. Loosen Hard stop Block Screw (3 x M4) and the Hard Stop Support Screw & Nut (M5) as Figure 12.

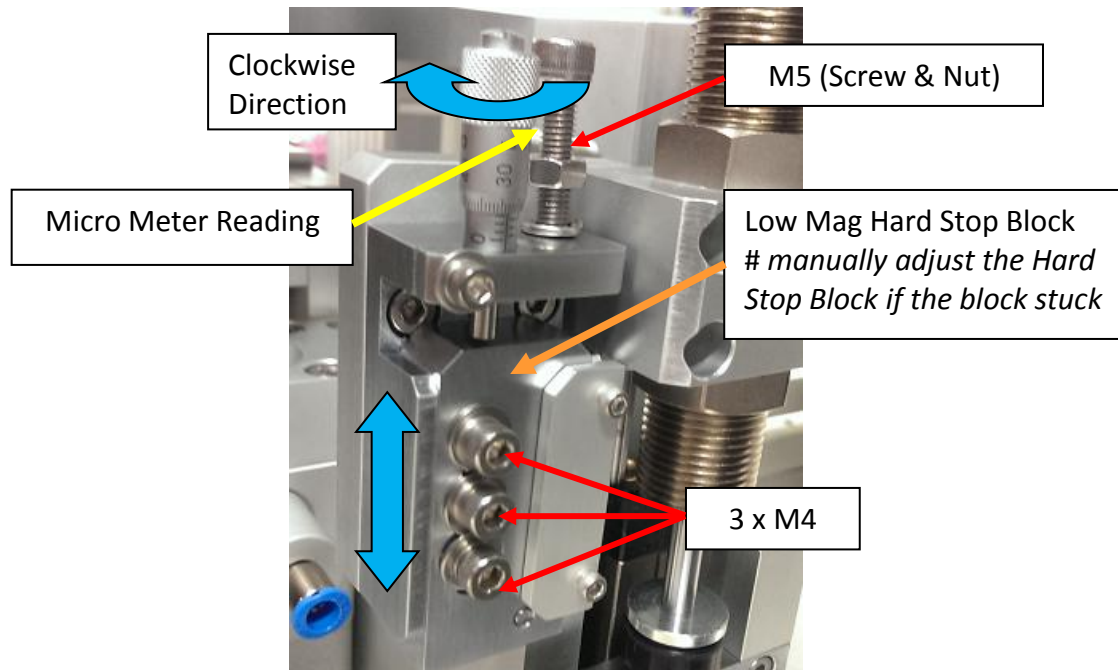


Figure 12 Low mag hard stop

15. Twist Micrometer with **Clockwise (CW)** will lower the tube and **Counter Clockwise (CCW)** higher the tube.

Depend on calculation step 6, 1 turn will be 40mil.

Need to manually adjust the Hard Stop Block if the block stuck by Block Holder until touch the Micrometer.

*# Spec tolerance **5.21 ~5.25***

16. Tighten the 3 Hard stop Block Screws (3 x M4) once adjustment done.

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

17. Go to Panel Handling tab, trigger tube cylinder to “Up” and “Down” position few times (manual exercise) and keep in Up position before proceed next step.

*Services → Subsystem Status & Control → Digital I/O → Inner Barrier Open Position → X-Ray Cylinder Up/Down Position
→ X-Ray Cylinder Up Position*

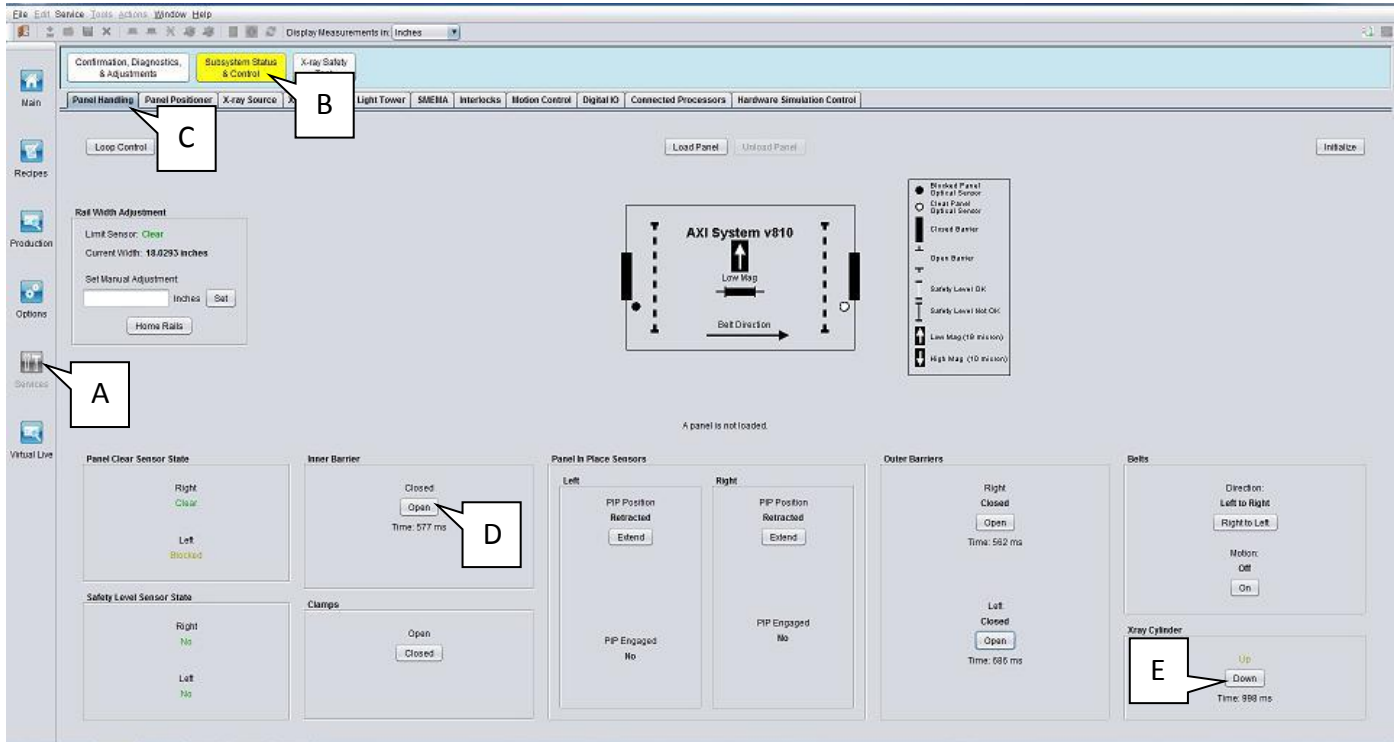


Figure 13 Panel Handling

18. Turn off main V810 air pressure regulator.



Figure 14

	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

19. Manually push the inner barrier shutter plate, ensure X ray tube bottom target is **NOT** contain any **friction** with Inner Barrier Shutter Plate as Figure 15.

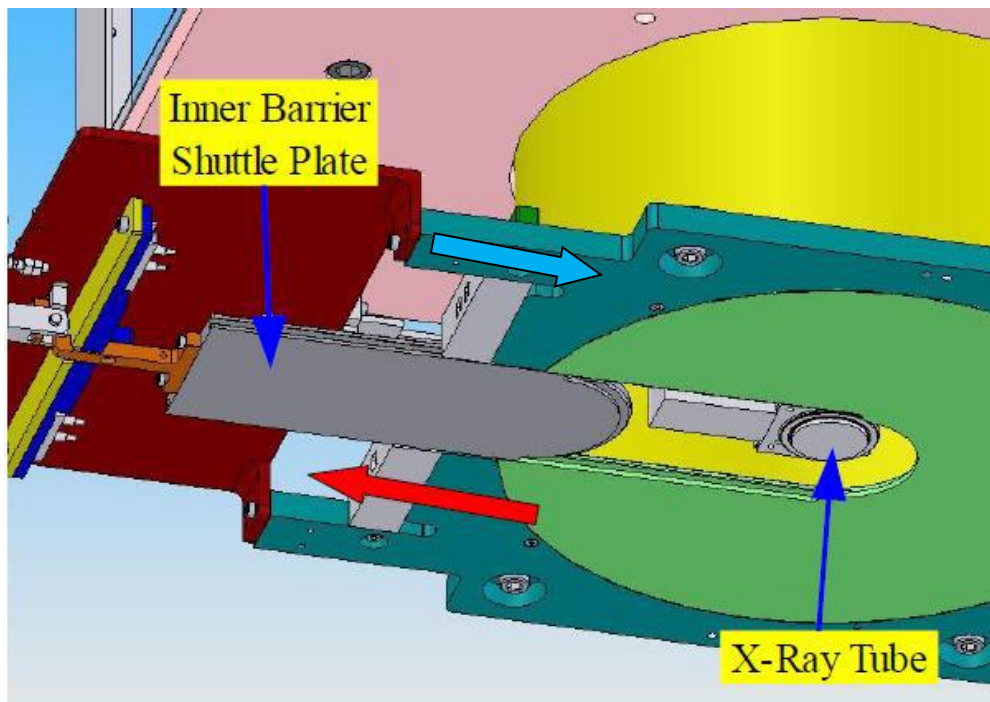


Figure 15 Tube target clearance checking

Remark,

Gap between shutter plate and tube Beryllium surface should be slightly as 0.5 mm, as Figure 16. If X-Ray tube too closes to Inner Barrier Shutter Plate, you may feel friction. Trigger cylinder down at Panel Handling and slightly adjust the Low Mag hard stop to higher position.

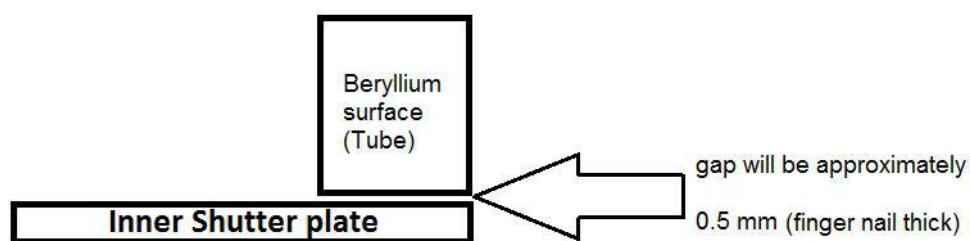


Figure 16

20. Turn On main V810 air pressure regulator and set incoming air source as 80 PSI.

	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

21. Go to Panel Handling tab, ensure tube cylinder in Up position, manually trigger Inner Barrier Open/Close few times, and check for the shutter plate movement as Figure 17. Ensure the movement is smooth, if not smooth, might be shutter plate friction with tube Beryllium surface. Trigger cylinder down at Panel Handling and slightly adjust the Low Mag hard stop to higher position.

Services → Subsystem Status & Control → Panel Handling → X-Ray Cylinder Down Position → Manually trigger Inner Barrier Open/Close

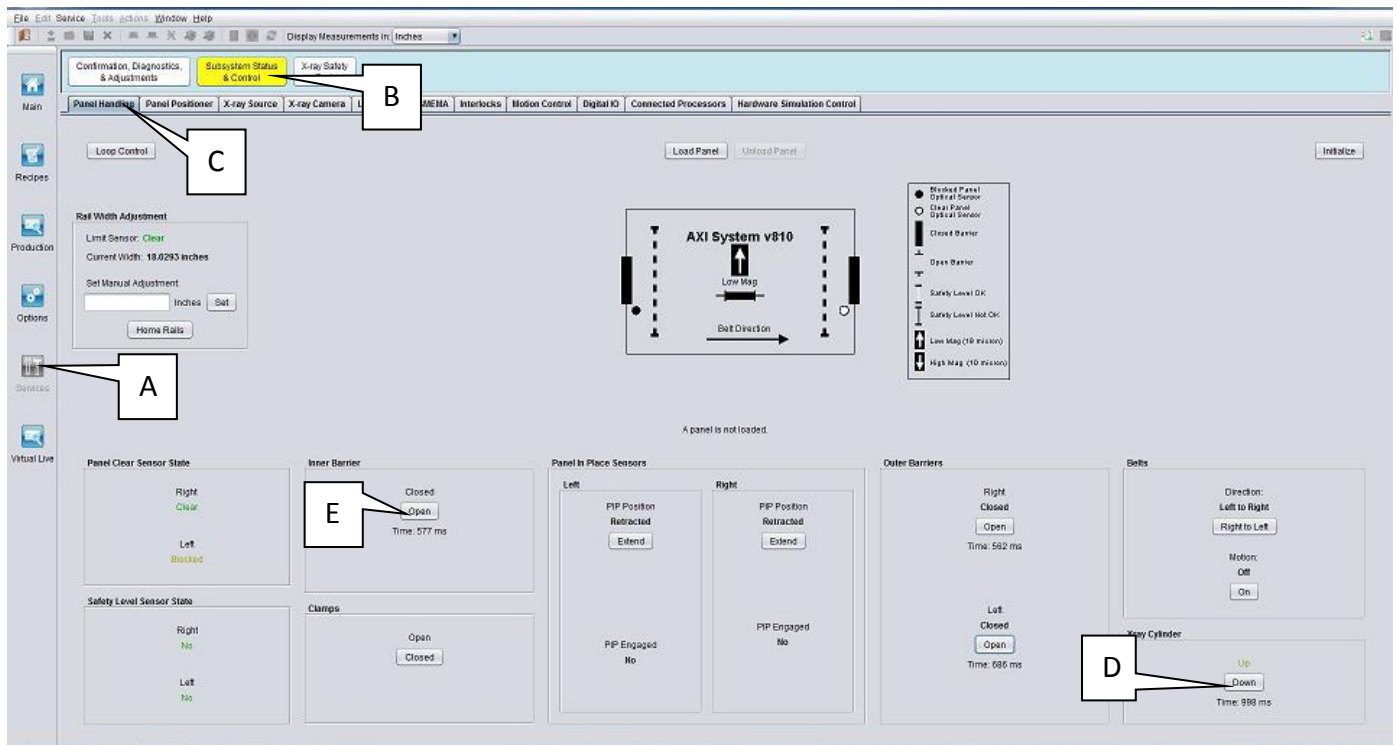


Figure 17 Panel handling

	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

22. With tube cylinder **Up** position, hand tight the Hard Stop Support Screw and Nut. Ensure the screw is touching the Hard Stop Block surface as Figure 18.

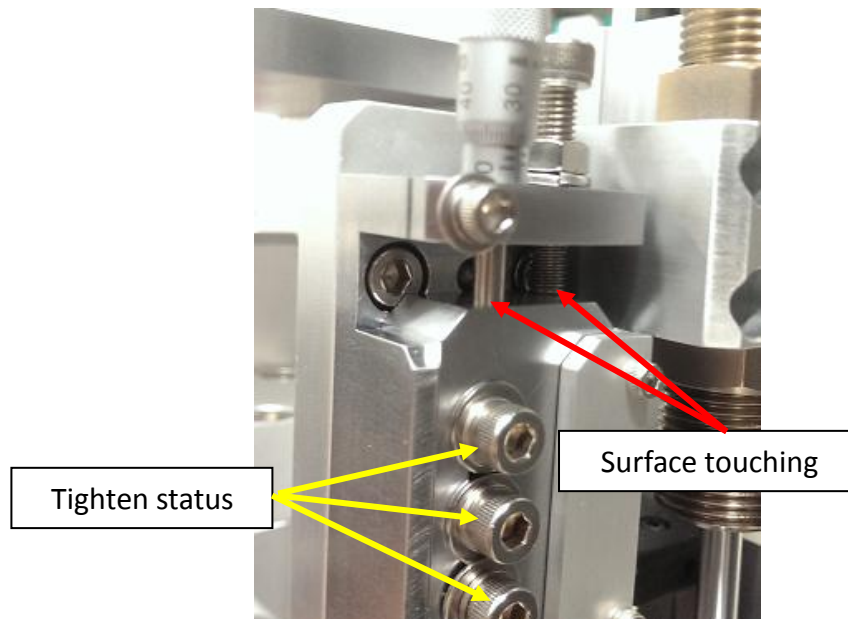


Figure 18 Low Mag hard stop

23. With tube cylinder up position, check for proximity sensor signal. The higher proximity sensor should emit Amber LED with tube up position as Figure 19. If the sensor **NOT** emitted, loosen the set screw and reposition the sensor from top to bottom until LED is turn On (Amber indicate).

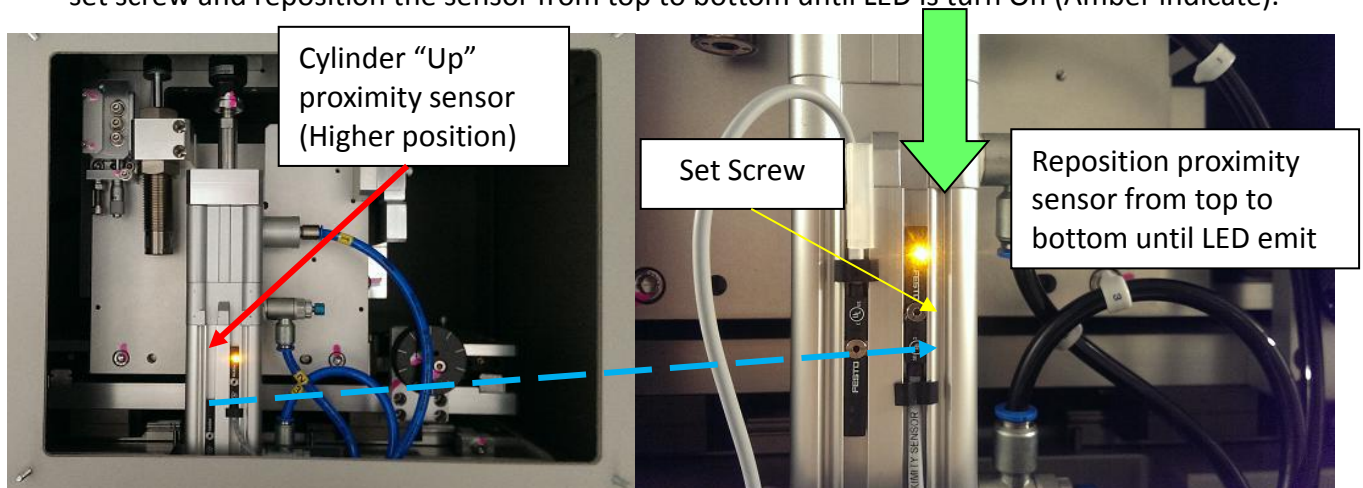


Figure 19

	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

27. High Mag Hard Stop and screws show as Figure 21.

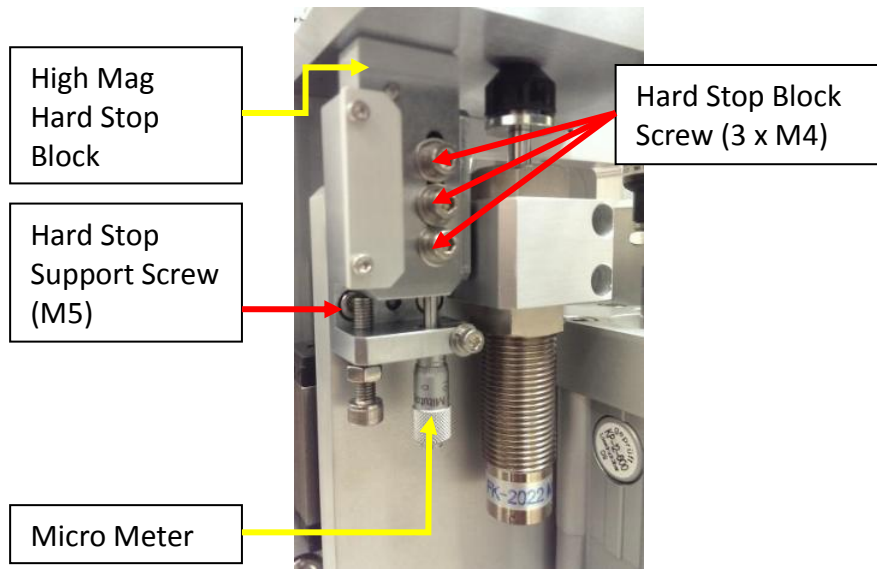


Figure 21 (New Rev)

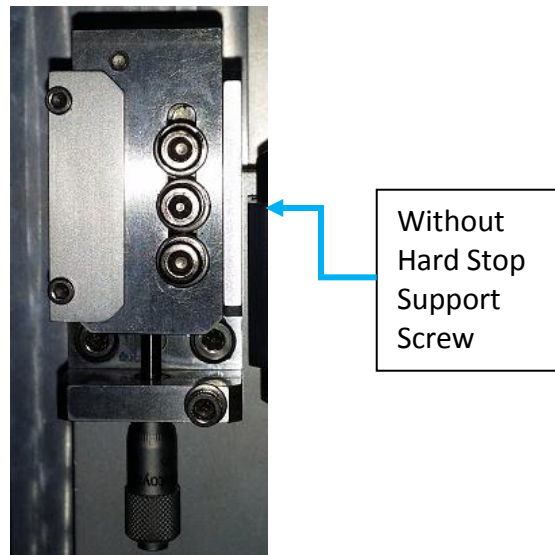


Figure 22 (Old Rev)

Remark,

If machine assembly with VM Old Rev as Figure 22, the difference will be no assembly Hard Stop Support Screw (M5), tube adjustment will be the same method.

	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

28. Remark, If wish to higher tube Z for High Mag adjustment, need to trigger tube in Down position

1st.

Go to Panel Handling tab, trigger tube cylinder to “Down” position 1st as Figure 23.

Services → Subsystem Status & Control → Panel Handling → Inner Barrier Open Position → X-Ray Cylinder Down Position

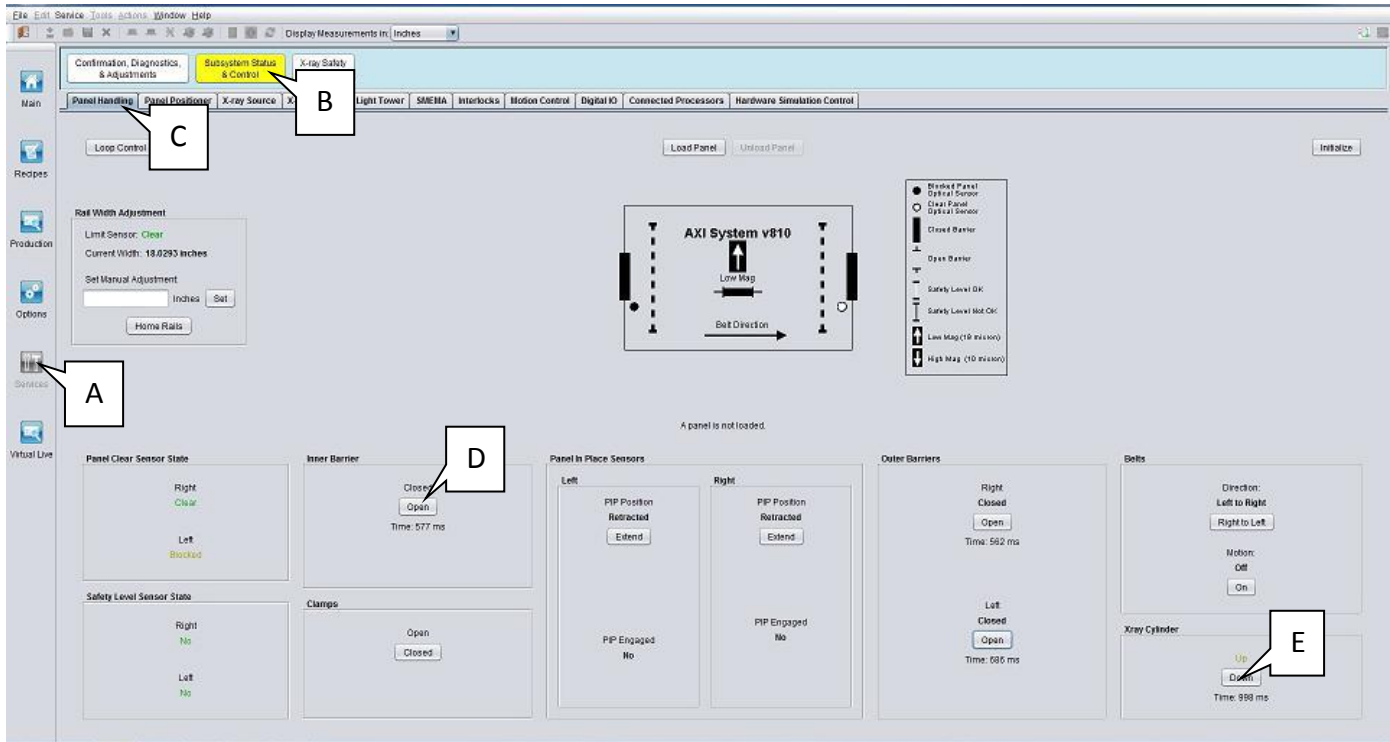


Figure 23 Panel Handling

29. Loosen Hard stop Block Screw (3 x M4) and the Hard Stop Support Screw & Nut (M5) as Figure 24.

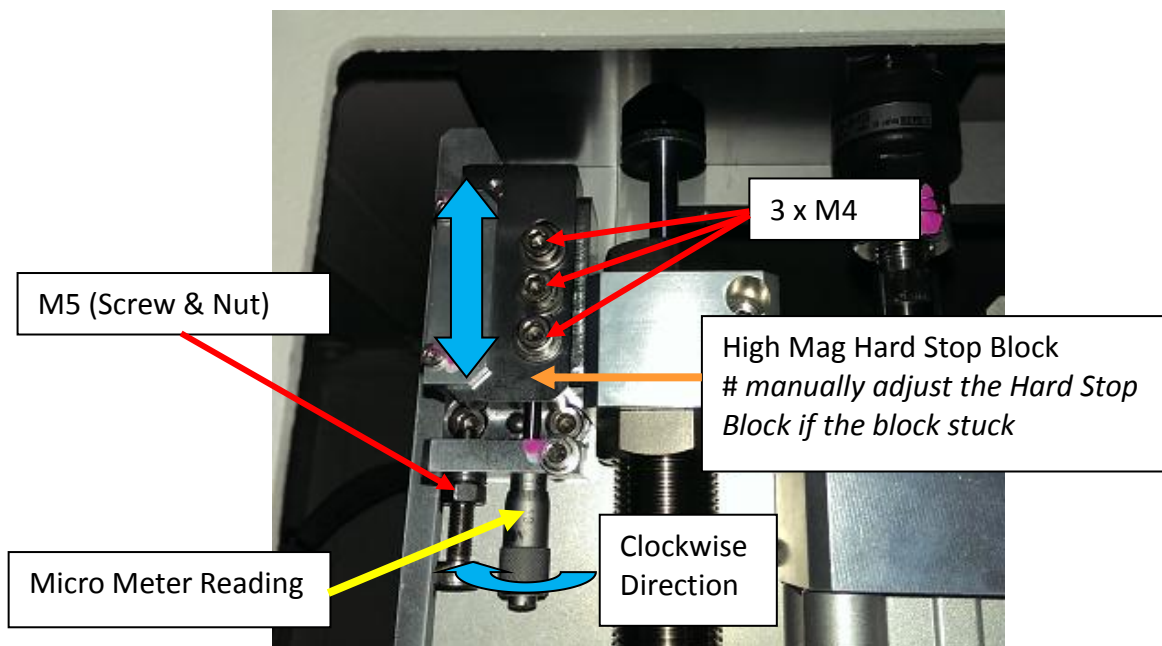


Figure 24 High mag hard stop

	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

30. Twist Micrometer with **Clockwise (CW)** will lower the tube and **Counter Clockwise (CW)** higher the tube.

Depend on calculation step 26, 1 turn will be 40mil.

Need to manually adjust the Hard Stop Block if the block stuck by Block Holder until touch the Micrometer.

*# Spec tolerance **9.05 ~ 9.09***

31. Tighten the 3 Hard stop Block Screws (3 x M4) once adjustment done.

32. Go to Panel Handling tab, trigger tube cylinder to "Up" and "Down" position few times (manual exercise) and keep in Down position before proceed next step.

Services → Subsystem Status & Control → Panel Handling → Inner Barrier Open Position → X-Ray Cylinder Up/Down Position → X-Ray Cylinder Down Position

33. Place the Ver Mag Jig (4XASS-A377) on the rail stage with Top Clearance Block face top as Figure 25.

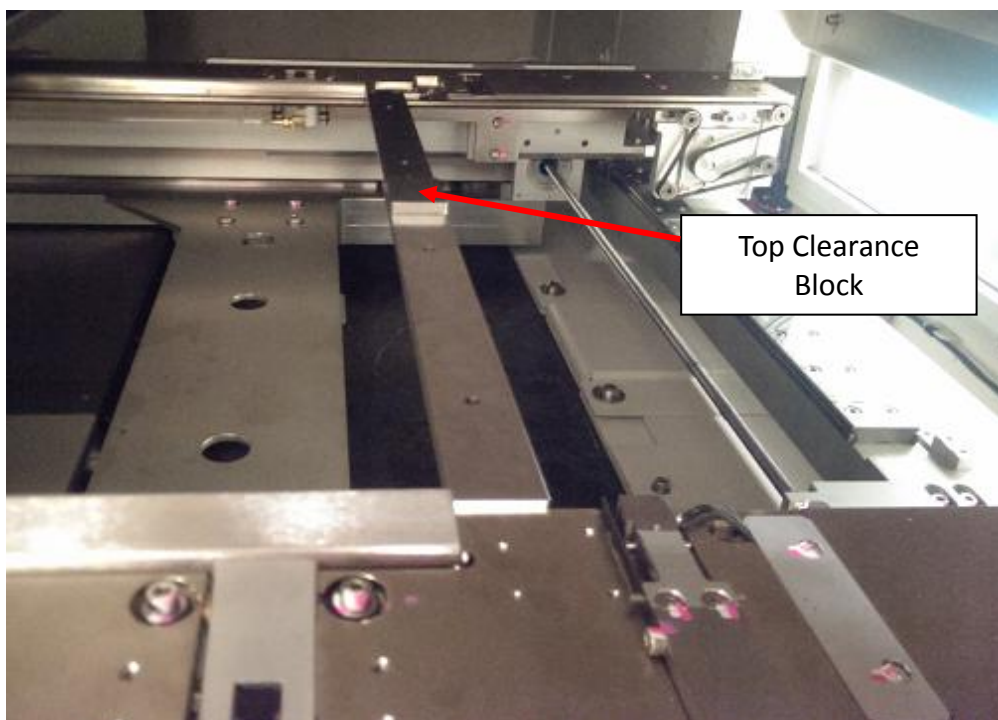


Figure 25 High Mag Clearance Checking

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

34. Move the XY stage pass thru the X-Ray tube bottom target and ensure Jig is not crash/friction with tube Figure 26 and the stage clamp blade as Figure 27. As usual the clearance between tube and Jig will be approximately 2~3mm. If X-Ray tube crashed, slightly adjust the High Mag hard stop to higher position.

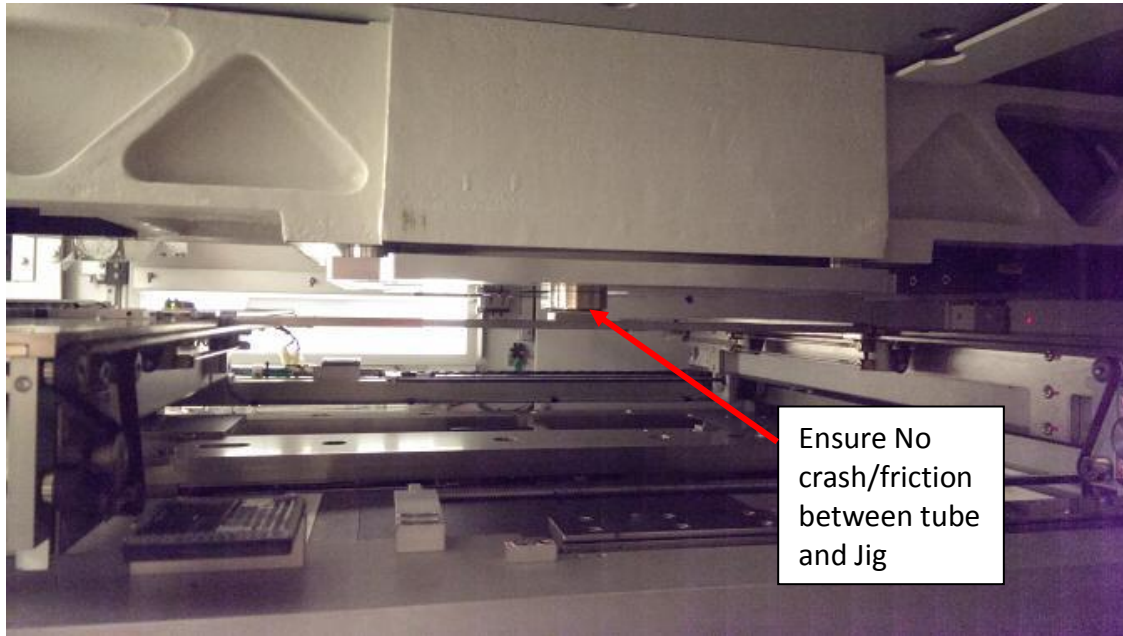


Figure 26 Clearance checking

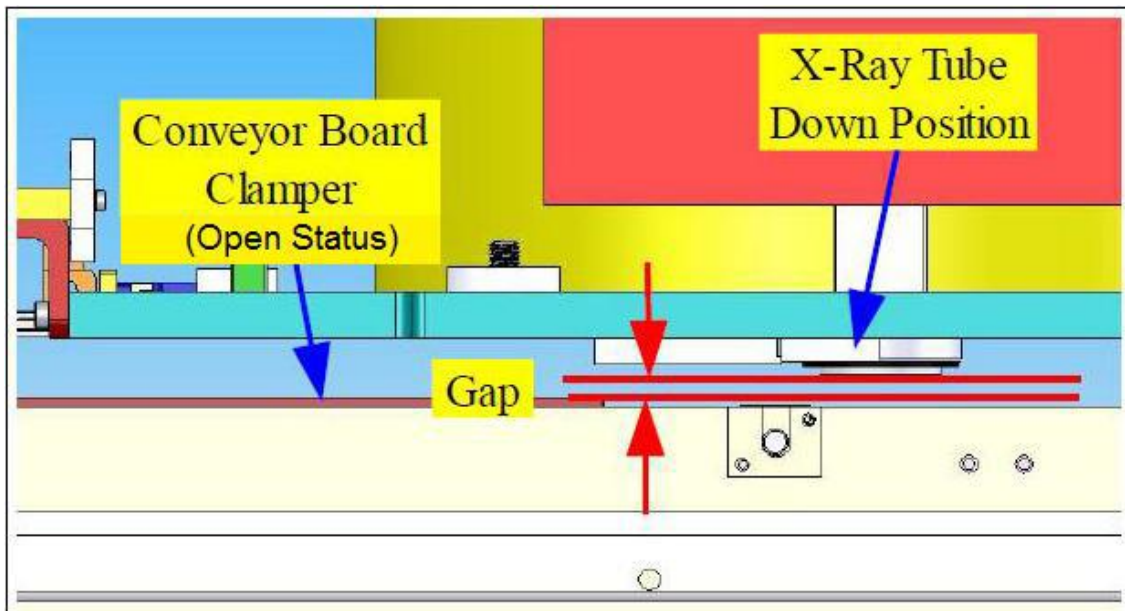


Figure 27 Clearance checking

	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

35. With tube cylinder **Down** position, hand tight the M5, Hard Stop Support Screw and Nut. Ensure the screw is touching the Hard Stop Block surface as Figure 28.

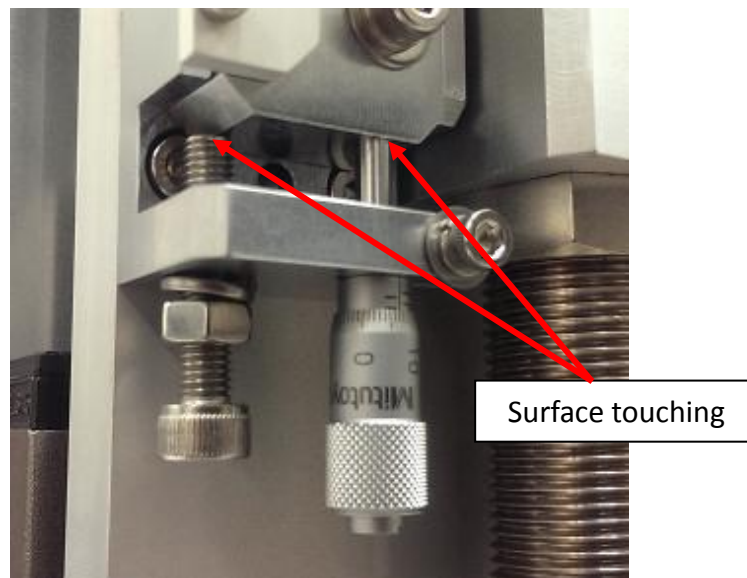


Figure 28 High Mag hard stop

36. With tube cylinder up position, check for proximity sensor signal. The higher proximity sensor should emit Amber LED with tube up position as Figure 29. If the sensor **NOT** emitted, loosen the set screw and reposition the sensor from bottom to top until LED is turn On (Amber indicate).

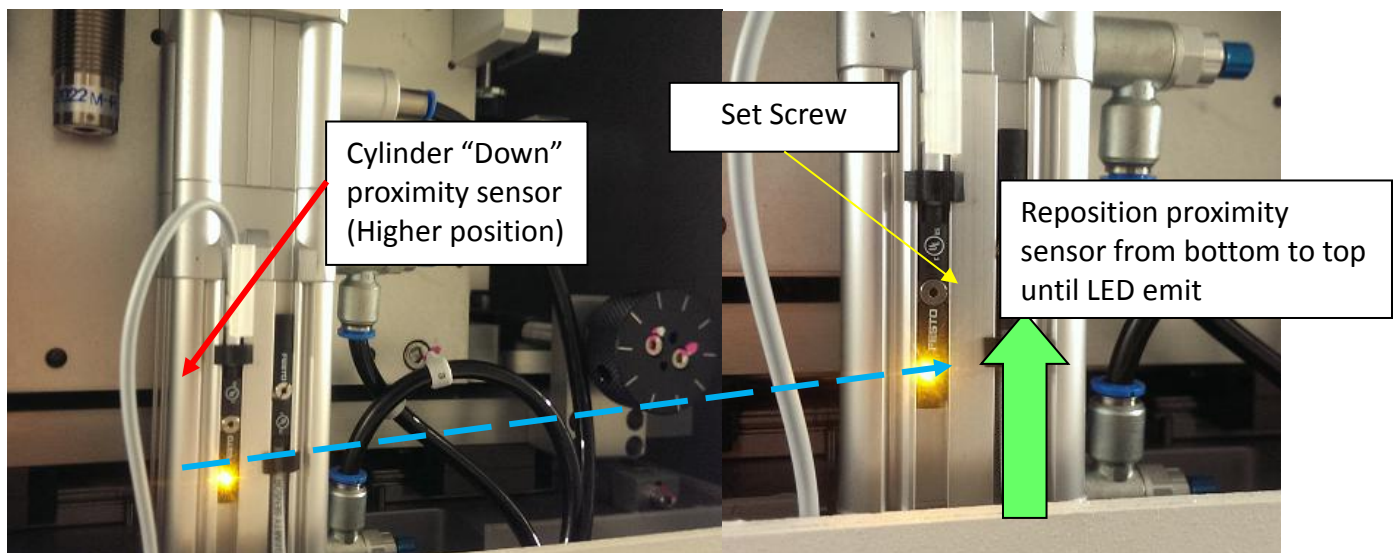


Figure 29 Proximity sensor adjustment

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 STD VM (19um and 11um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

37. Go to Panel Handling tab, trigger tube cylinder to “Up” position as figure 30.

Services → Subsystem Status & Control → Panel Handling → Inner Barrier Open Position → X-Ray Cylinder Up Position.

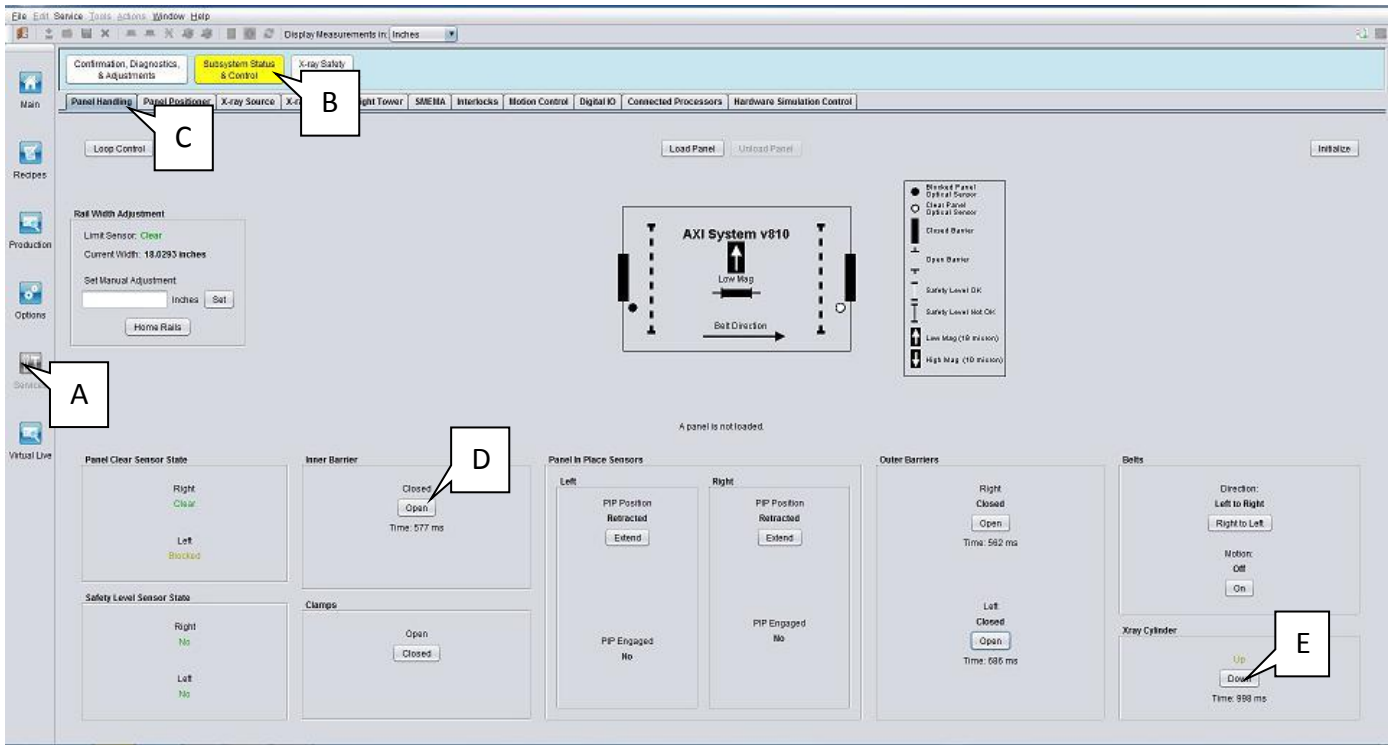


Figure 30 Panel Handling

38. Assembly the U-Shield Front Cover with fastener 4 x M5 nuts.

39. Remove all unnecessary tools and jig from machine.

Start-up machine as usual.

40. Run full CDnA 5 loop and verify each loop magnification log file (Low and High Mag), ensure result is within spec.

Low Mag Spec tolerance 5.21~5.25

High Mag Spec tolerance 9.05~9.09